



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (03/02/2025 – 09/02/2025)

WEEK 1

PROJECT NAME	CHANGE DETECTION
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PROJECT DESCRIPTION :

This project aims to detect changes between satellite images captured at different times using Deep Learning and Computer Vision techniques.

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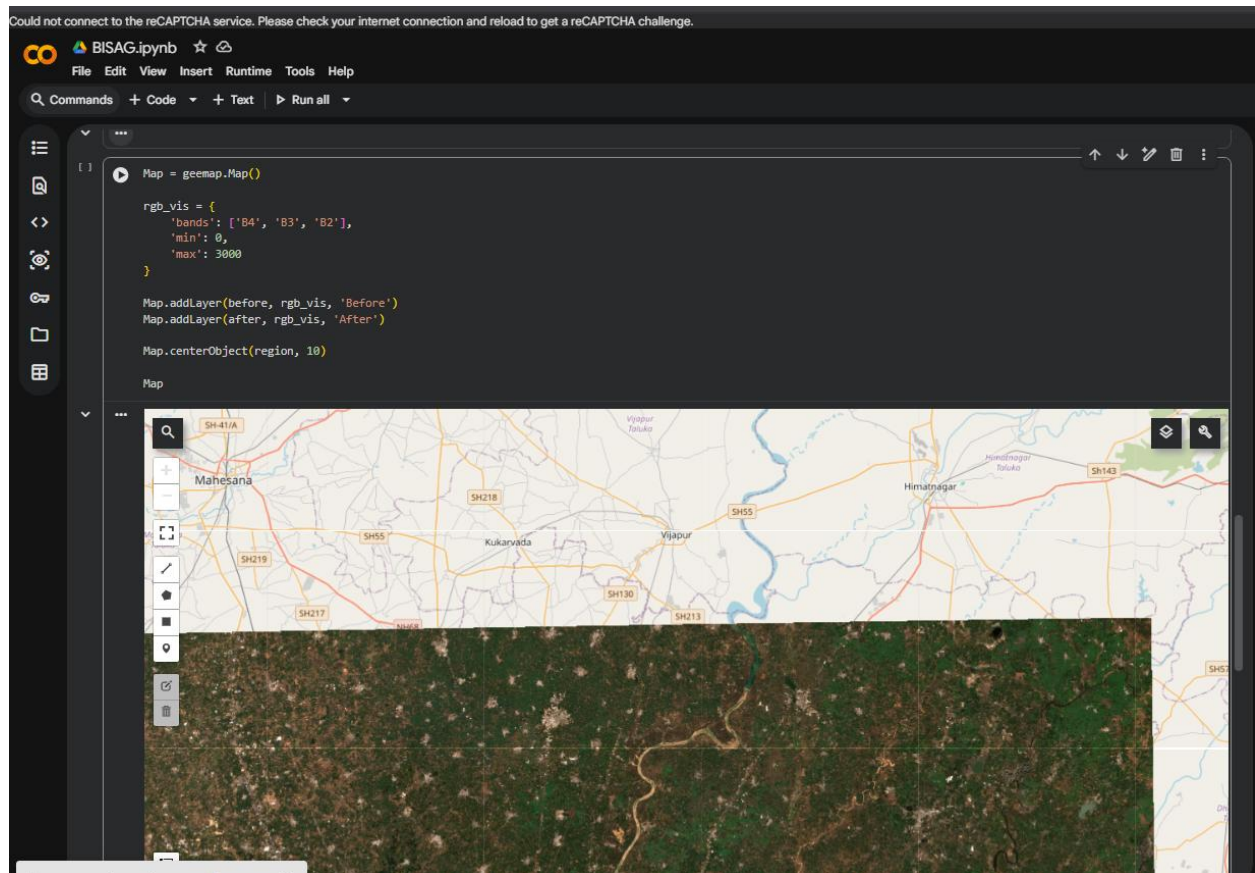
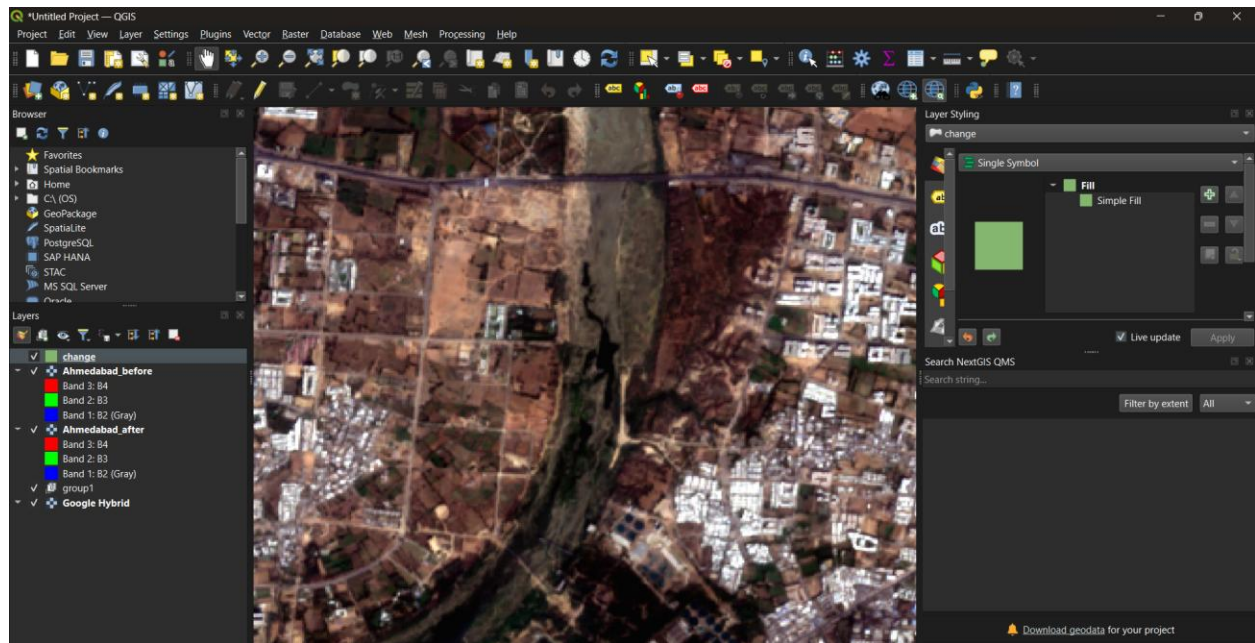
18/05/2026	• Overview of the projects and meeting the project guide.
19/05/2026	• Assigned with the project title “CHANGE DETECTION”.
20/05/2026	• Gained basic knowledge of the various models designed for the Satellite image change detection. • Basic models like FCSN,FC-EF, Siamese etc.
21/05/2026	• Moved towards the transformer based models like STANet, BIT etc.
22/05/2026	• Exploring the working with Google earth engine. • Creation of dataset with GEE and QGIS.
23/05/2026	Holiday (2 nd Saturday)
24/05/2026	Holiday (Sunday)

Next Week Plan	Planning to work with the same work flow (GEE+QGIS) and along side exploring other alternatives which can be useful to complete our project.
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REFERENCE:

- <https://github.com/likyoo/open-cd/tree/main/configs/fcsn>
- https://developers.google.com/earthengine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED

SCREEN SHOTS:



Files

main

Go to file

- base
- ban
- bat
- cgnet
- changeformer
- changer
- changestar
- common
- fcnn
 - README.md
 - fc_siam_diff_256x256_40k_invertedpy
 - fc_siam_conc_256x256_40k_invertedpy
 - fc_siam_diff_256x256_40k_invertedpy
- general_scd
- hanet
- hsv
- lightnet
- mskd
- svanet
- starc
- trycd
- trycd_v2
- ttp
- docs
- opencd
- projects
- requirements
- resources

open-cd / config / fcnn /

fc_siam_diff_256x256_40k_invertedpy

release Open-CD v1.0.0

3 years ago

README.md

FCSN

[Fully Convolutional Siamese Networks for Change Detection](#)

Introduction

[Code Snippet](#)

Abstract

This paper presents three fully convolutional neural network architectures which perform change detection using a pair of coregistered images. Most notably, we propose two Siamese extensions of fully convolutional networks which use heuristics about the current problem to achieve the best results in our tests on two open change detection datasets, using both RGB and multispectral images. We show that our system is able to learn from scratch using annotated change detection images. Our architectures achieve better performance than previously proposed methods, while being at least 500 times faster than related systems. This work is a step towards efficient processing of data from large scale Earth observation systems such as Copernicus or Landsat.

(a) FC-EE

(b) FC-Siam-conc

(c) FC-Siam-diff

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